Soban Nasir Lone

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EDUCATION

Technical University of Munich, Germany — Jan 2025 - Current
Doctoral candidate, Chair of Transportation Systems Engineering (TUM ED)

Indian Institute of Technology, Delhi, India — Aug 2022 - Oct 2024
Master's by Research in Applied Mechanics — Cumulative GPA: 8.906/10

National Institute of Technology, Srinagar, India — Aug 2017 - July 2021
Bachelor's of Technology in Civil Engineering — Cumulative GPA: 8.435/10

CAREER BREAK

Preparation for entrance examination - Graduate Aptitude Test in Engineering 2022 — Aug 2021 - March 2022
Secured admission to the Indian Institute of Technology, Delhi

Up-skilling and preparation for Master's degree — April 2022 - June 2022
Spent 2-3 months before the master's program learning Python, introductory machine learning, and strengthening fundamentals in Strength of Materials, Linear Algebra, and Calculus.

Up-skilling and preparation for PhD — Nov 2024 - Dec 2024
Spent 2 months before the start of PhD learning Graph neural networks

PUBLICATIONS

Journal & Conference Publications

- **Lone, S. N.**, De, S., & Nayek, R. (2026). α-VI DeepONet: A prior-robust variational Bayesian approach for enhancing DeepONets with uncertainty quantification. *Computer Methods in Applied Mechanics and Engineering*, 449, 118552. https://doi.org/10.1016/j.cma.2025.118552
- Yu, T., Kim, J., **Lone, S. N.**, Abouelela, M., & Antoniou, C. (2025). Scalable traffic prediction via data reduction with coreset selection. In *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 2069–2076). Gold Coast, Australia. https://doi.org/10.1109/ITSC60802.2025.11423766

Preprints (Under Review)

- Noor, T., **Lone, S. N.**, Ramana, G. V., & Nayek, R. (2025). A recursive Bayesian neural network for constitutive modeling of sands under monotonic loading. *arXiv preprint arXiv:2501.10088*. https://arxiv.org/abs/2501.10088
- **Lone, S. N.**, Abouelela, M., Yu, T., Kim, J., & Antoniou, C. (2025). Transfer learning in spatiotemporal graph neural networks for traffic forecasting: Taxonomy, analysis, and future directions. *SSRN*. https://dx.doi.org/10.2139/ssrn.5933134

ACADEMIC EXPERIENCE

Technical University of Munich, Germany
PhD Candidate, Chair of Transportation Systems Engineering — Jan 2025 - Present
Supervisor: *Prof. Dr. Constantinos Antoniou*
Co-Supervisor: *Prof. Dr. Jiwon Kim*
Project: **Model Interpretation and Data-centric Modeling for Advanced traffic Prediction (MINDMAP)**

Indian Institute of Technology, Delhi — Aug 2022 - Oct 2024
Student Researcher - System Monitoring, Identification, and Control Research (SMICR) Group
Supervisor: *Dr. Rajdip Nayek*
Thesis: **Applying Generalised Variational Inference for Robust Uncertainty Quantification in Bayesian DeepONets and Soil Constitutive Modelling**

Projects:

Bayesian neural networks (BNNs) for constitutive modelling — Oct 2023 - Present

- Researched, analysed, and gained expertise in BNNs for uncertainty quantification in deep learning.
- Applied BNNs for constitutive modelling of soil trained on time-series stress-strain data.

Physics Informed Neural Networks (PINNs) — Mar 2023 - May 2023

- Modelled the deformation of a 2D bar subjected to in-plane loading and compared effects of using different forces.
- Implemented codes for forward and inverse problems of a structural dynamical system using PINNs in TensorFlow.

Bayesian Physics Informed Neural Networks (B-PINNs) Dec 2023 - Jan 2024

- Extended the deterministic PINN architecture to include a Bayesian neural network for uncertainty quantification.
- Incorporated a methodology that remedies prior-misspecification for Bayesian neural networks and thus improved predictive performance.

Parameter Estimation of Dynamical Systems Jul 2023 - Oct 2023

- Developed a state-space model to estimate parameters and quantify uncertainty in a structural dynamical system using Markov chain Monte Carlo (MCMC), Kalman Filter and Variational Inference.
- Analyzed advantages, limitations, and computation times of the algorithms for accuracy, convergence, and efficiency.

National Institute of Technology, Srinagar
Undergraduate Research, Department of Civil Engineering Dec 2020 - Jun 2021
Supervisor: *Prof. F.A. Mir*
Final year project: **Site-Specific Landslide Analysis and Correction**

- Collected data, including soil samples and site geography, as part of a comprehensive project on Landslide Analysis.
- Conducted in-depth soil data analysis, utilizing experimental methods and simulations (GeoStudio-GeoSlope) to gain insights into properties and characteristics of the site and propose possible corrections.

Indian Institute of Technology, Delhi Jul 2020
Research Intern, Department of Civil Engineering
Project: **Microtremor and multichannel analysis of seismic waves for microzonation of an earthquake prone region**

- Microtremor and multichannel analysis of seismic waves for microzonation of an earthquake prone region

RESEARCH INTERESTS

- Traffic Prediction using Graph Neural Network-based Models
- Transfer Learning of Traffic Prediction Models
- Explainable AI; Mechanistic Interpretability
- Bayesian Neural Networks for Uncertainty Quantification using Variational Inference.
- Operator Learning frameworks for solving ODEs and PDEs.
- Scientific Machine Learning applications in Civil Engineering.

SELECTED COURSES

Master's Courses

- Deep Learning for Mechanics
- Introduction to Machine Learning
- Decision Theory and Design Optimisation
- Engineering Mathematics and Computation
- Dynamics

Bachelor's Courses

- Computer Programming (C Programming)
- Mathematics (Linear Algebra, Statistics and Probability, Calculus)
- Traffic Engineering
- Highway Engineering

AWARDS AND ACHIEVEMENTS

- Secured a percentile of 98.5 (out of 100,043 candidates) in the **G**raduate **A**ptitude **T**est in **E**ngineering (**GATE**) (Civil Engineering) 2022 for admission to graduate level programmes to the IIT's and IISc.
- Obtained a percentile of 97 (out of 124,000 candidates) in JEE MAINS 2017 for admission to undergraduate level programmes to the IIT's, NIT's and IIIT's.
- **COLOURS (Football)** by Board of Sports Activities (BSA), IIT Delhi (2022-23) for outstanding achievements in sports.
- Post Graduate Scholarship by the Ministry of Human Resource Development (MHRD), Govt. of India for Academic years 2022 to 2024.

TEACHING EXPERIENCE

Technical University of Munich, Germany

- Lecturer for the course Statistical Learning and Data Analytics for Transportation Systems. Summer 2026.
- Lecturer for the course Urban Operations Research for Transportation Systems. Summer 2026.

- Lecturer for the course Transportation Economics. Winter 2025.

Indian Institute of Technology, Delhi

- Teaching Assistant for the course Engineering Mechanics (APL100). Fall 2022
- Teaching Assistant for the course Finite Element Methods (AML706). Spring 2023
- Teaching Assistant for the course Applied Mathematics for Engineers (APL101). Fall 2023
- Teaching Assistant for the course Machine Learning for Mechanics (APL405). Spring 2024

SKILLS

- **Programming:** Python, R, C Programming, MATLAB
- **Frameworks:** TensorFlow, TensorFlow Probability, PyTorch, PyMC3
- **Others:** GitHub, LaTeX, Matplotlib, Scipy

EXTRACURRICULARS

- Explored AI Safety concepts as a fellow at the AI Safety Careers Fellowship by FAR AI, Inc. at IIT, Delhi in 2023.
- Represented Indian Institute of Technology, Delhi at the Inter-IIT Sports Meet (the annual Olympic-style sports tournament of the 23 Indian Institutes of Technology) in 2022 and 2023 as a part of the institute football team, securing 3rd and 4th positions in the overall standings respectively.
- Participated in Udghosh (the Annual Sports Festival of IIT, Kanpur) in 2022 and 2023, winning Gold (Football) in 2023.
- Captained my Hostel team to a podium finish in the Inter-Hostel football tournament (2023) organised by IIT Delhi.
- Participated in the Inter-NIT Football tournament held at National Institute of Technology, Rourkela in 2020.

REFERENCES

Dr. Radjip Nayek
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Scholar Profiles: Group (SMICR) — Google Scholar — LinkedIn

Dr. Souvik Chakraborty
Assistant Professor, Department of Applied Mechanics, Indian Institute of Technology, Delhi, India
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Scholar Profiles: Group (CSCCM) — Google Scholar — LinkedIn

Dr. Falak Zahoor
Assistant Professor, Department of Civil Engineering, National Institute of Technology, Srinagar, India
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Scholar Profiles: Google Scholar